Feature Analysis of Transformer Model for The Disturbance Storm Time Index

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Abstract

In space weather research, the measurement of disturbance storm time (DST) index is valuable. It is useful in characterizing the properties like size and intensity for a geomagnetic storm. During storms, the reduction of Earth's magnetic field is indicated by a negative DST index. These index values can be forecasted based on the solar wind parameters available through the NASA Space Science Data Coordinated Archive with help a transformer model called DST Transformer which is a combination of multi-head attention layer and Bayesian inference. Though the performance of the model is good in terms of the root mean square error and R-squared, it is important to understand the importance of features utilized by the model to achieve the performance for better understanding of the model. In this paper, we present feature analysis of the DST Transformer to understand model performance based on the solar parameters using SHAP and LIME. SHAP is an explainable AI tool which uses the game theoretic approach to explain the output of the model. The SHAP results explain that the performance of model is mostly based on certain set of parameters instead of the complete set. LIME (Local Interpretable Model-Agnostic Explanations) is a technique in explainable AI that approximates the local model with any black box model to explain prediction. LIME results show that there exists some difference in behaviour of the model bet positive and negative samples. These results can help us to tune the model by removing the features that do not show any significant effect. Additionally, this paper also presents feature analysis based on the attention scores of multi-head attention layer of DST transformer.

Introduction

Geomagnetic study is an essential part of space weather research due to its effect on Earth which includes causing damage or disturbance in networking and power grid transmission systems. This study is carried out with the help of solar activity which includes parameters such as interplanetary magnetic field (IMF), total electric field, solar wind speed and plasma temperature. The disturbance storm time (DST) index is an important factor useful for characterizing geomagnetic storm properties. As DST value decreases the intensity of the storm increases.

The DST values can be forecasted before with help of the solar activity parameters which allows us to study and take necessary precautions if needed. There are many techniques and models available for DST predictions like differential equations model according to Burton, McPherron, and Russell (1975), time delay ANN from Gleisner, Lundstedt, and Wintoft (1996), ANN with swarm optimization technique from Lazzús et al. (2017), etc. There are also models that go beyond single point approach like probabilistic forecasting according to Chandorkar, Camporeale, and Wing (2017).

Due to the presence of many models, it is essential to understand how these models work and what features support the model to achieve its performance. A model is fed with many features but not all features contribute to the final model. These features can be identified based on different model explanation techniques. The most effective techniques or methods are Explainable AI based methods. Explainable AI is a framework useful in interpreting the outcomes of the model and understanding of the feature importance in the predictions.

In this paper, we present feature analysis of the DST Transformer to understand model performance based on the solar parameters. SHAP is an explainable AI tool introduced by Lundberg and Lee in 2016. It is based on the game theory on optimal Shapley values. Shapley values explain the average contribution of features in all possible interactions. LIME (Local Interpretable Model-Agnostic Explanations) is a technique in explainable AI that approximates the local model with any black box model to explain prediction. LIME attempts to understand the model by observing the prediction change based on varying input samples.[1] The model used for this analysis is a transformer model based on Bayesian deep learning approach from Abduallah et al. (2022). Along with SHAP and LIME, we present feature analysis using attention scores of multi-head attention layer of the DST transformer. These attention scores provide a different view of the features based on the attention layer.

Data

Data used in this paper is the same data used to train and test the DST transformer model from the paper by Abdual-

[1] https://xai.arya.ai/wiki/lime

lah et al. (2022). The data was originally provided by the NASA Space Science Data Coordinated Archive.[2] The time period for the DST index data is in between January 1, 2010, and November 15, 2021, with hourly average as time resolution. The dataset is divided into two parts: training set with 102,976 records from January 1, 2010, to September 30, 2021 and test set with 1104 records from October 1, 2021 to November 15, 2021. Each record in the train set contains eight values in which seven values are solar wind parameters and one is label value.[3]

Methodology

Architecture of the DSTT Model Used

The architecture of the DST transformer used for this analysis is shown in figure 1. This model uses multiple layers created using the tensorflow keras framework.[4] The model accepts a sequence of non-overlapping records which are passed through a one dimensional convolution neural network (Conv1D) with 32 kernels of size 1 each. The output of Conv1D layer is then passed to a long short-term memory (LSTM) layer with 250 units which passes its results to a multi-head attention layer (Vaswani et al. 2017).

A custom attention is added to instruct layers to focus on the critical information regarding the input sequence and capture its correlation with the output by computing weighted sum of data sequence. Then a dense variational layer (DVL) (Tran et al. 2019) with 10 neurons is added that uses variational inference (Blei, Kucukelbir, and McAuliffe 2016) to estimate posterior over the weights along with multiple dense and dropout layers

Feature Analysis

For this analysis we use SHAP Kernel Explainer or Kernal SHAP[5] to explain the results of the models. First, the explainer is initialized with the model and the sample data of size 100. The sample data is obtained by applying SHAP kmeans clustering on the train data of size 104075 which outputs 100 clusters center representing the data. This step is carried out to resample the data which acts as a reference for SHAP explainer.

Then, Shapley values are generated by passing the sample test data of size 1098 through shap_values() method of the explainer. These Shapley values are used for different plots. For our analysis we use three plots namely Summary, Decision, and Dependence plot.

For LIME analysis, we use LIME tabular explainer [6] to interpret the model locally. The explainer is initialized using complete training data with mode set to regression. Then two instances are selected one being positive and one being negative from the test set. Now each instance is fed to the explainer, and explanations for the concerned predictions are obtained. For attention plots, attention weights are extracted from the attention layer and based on the average of the weights heat maps are generated.

[2] https://nssdc.gsfc.nasa.gov

[3] https://journals.flvc.org/FLAIRS/article/view/130564/133915

[4] https://www.tensorflow.org

[5] https://shap-lrjball.readthedocs.io/en/latest/generated/shap.KernelExplainer.html

[6] https://lime-ml.readthedocs.io/en/latest/lime.html

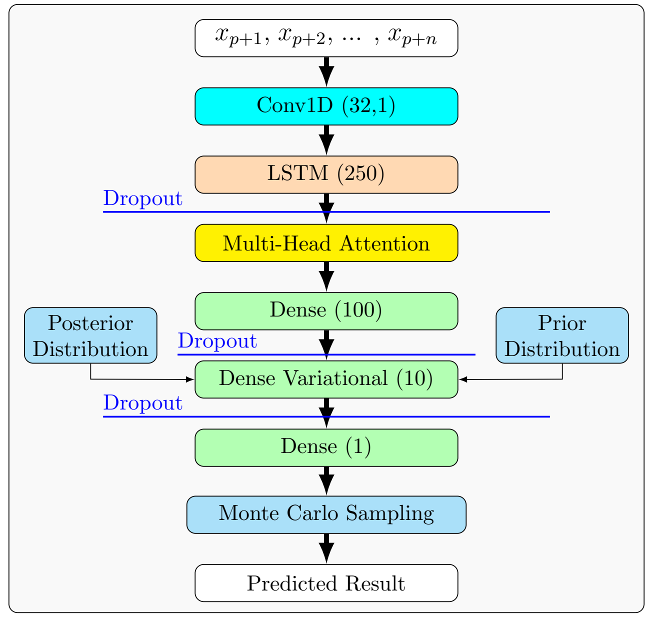

Figure 1: Architecture of our DST Transformer (DSTT)[3].

Results

SHAP Analysis

The SHAP values are plotted using three different graphs namely Summary, Dependency and Decision plot. Summary plot explains relationship of feature importance with feature effect. Dependency plot visualizes feature and prediction of model relation . Decision plot describes how a model arrives at its decision.

SHAP Summary Plot Inference

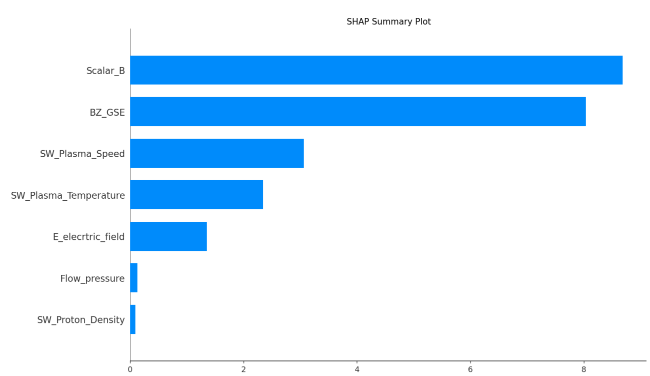

Figure 2: SHAP Summary Plot for DST Transformer

The summary plot describes the importance of each feature on model's output.

Figure 2 represents the summary plot for the seven parameters used for training the model. The parameters include Scalar_B, BZ_GSE, SW_Plasma_Temperature, SW_Proton_Density, SW_Plasma_Speed, Flow_pressure, E_electric_field.

According to the plot, Scalar_B is the most important feature followed by BZ_GSE. SW_Plasma_Speed and SW_Plasma_Temperature are of moderate importance. E_electric_field is less impactful than SW_Plasma_Temperature but still seems to be relevant to the model. SW_Plasma_Speed and Flow_pressure are least important features to the model

SHAP Dependency Plot Inference

Figure 3 show dependency plot illustrating relationship between the feature Scalar_B and its SHAP values with interaction effect of the feature BZ_GSE . Based on the interaction scores of different features with respect to Scalar_B, BZ_GSE is the highest interacting feature with a value of -0.9556. Here negative sign indicates that Scalar_B and BZ_GSE are negatively correlated.

The plot contains two clusters one at Scalar_B value zero and other at 1000. In plot, the interaction effect of BZ_GSE on Scalar_B feature and its SHAP value increases with increase in Scalar_B value. SHAP values tend to be more positive when Scalar_B value is around zero and tend to be more negative when Scalar_B value is around 1000 indicating a negative efect on model output. The effect of BZ_GSE on Scalar_B seem to be positive. The cluster around zero value is blue indicating presence of low BZ_GSE values. The cluster around 1000 is red indicating presence of high BZ_GSE values.

SHAP Decision Plot Inference

According to figure 4, the SW_Plasma_Speed feature and SW_Plasma_Temperature provide a significant negative impact on the output of the model. BZ_GSE and Scalar_B also reduce the model output but with a minor negative impact.

Flow_pressure, E_electric_field and SW_Proton_Density provide a small positive impact, contradicting previous negative impact and increasing model output slightly

LIME Analysis for Negative Instance

Two different plots namely, summary and decision plots, are used to explain the LIME results.

Figure 5 represents the summary plot of LIME explanations obtained using a negative sample from the test data. According to the graph, SW_Plasma_Temperature has the highest negative impact on the model output in case of the negative instance followed by the features SW_Plasma_Speed and BZ_GSE which show some negative impact. The feature Scalar_B so some positive impact on the model output balancing the the negative impact of previous features to some extent. Rest of the features show some positive impact which are not significant enough compared to previous features.

Figure 6 shows a decision plot of LIME explanations obtained using a negative sample from the test data. In this plot, SW_Plasma_Temperature is considered the starting point with highest negative contribution. As we move forward, the negative impact slowly decreases and with negligible change between the features SW_Plasma_Speed and BZ_GSE. Then it reaches highest possible position for the feature Scalar_B indicating significant positive impact from Scalar_B and slowly decreasing for other features but still remaining in positive zone.

LIME Analysis for Positive Instance

Figure 7 is a summary plot of LIME explanations obtained using a positive sample from the test data. According to the graph, BZ_GSE has the highest positive impact on the model output in case of the positive instance followed by the feature SW_Plasma_Temperature which show some significant positive impact. The features SW_Plasma_Speed and Scalar_B show some nearly equal positive impacts on the model output but with less impact than SW_Plasma_Temperature. The rest of the features show a somewhat lesser positive impact.

Figure 8 illustrates a decision plot of LIME explanations obtained using a positive sample from the test data. In this graph, BZ_GSE is referred as the starting point with highest positive contribution. As we move forward, the positive impact steadily decreases until the feature SW_Plasma_Speed. After that, it decreases slowly, with negligible changes between the features SW_Plasma_Speed and Scalar_B. The decrease continues until the final feature, Flow_pressure where the value is nearly zero, indicating that features after Scalar_B might not provide any significant impact.

Attention Head based Feature Analysis

Figure 9 depicts the visualization of attention map with four heads on the features BZ_GSE and Scalar_B. Each head captures different patterns between the features.

According to Head 1, Scalar_B has high attention weight when queried with both BZ_GSE and Scalar_B whereas BZ_GSE has high attention weight when queried with Scalar_B and low attention weight when queried with BZ_GSE. In case of Head 2, Scalar_B results are similar to Head 1 but BZ_GSE has low attention weight when queried with both BZ_GSE and Scalar_B. For Head 3, Scalar_B has high attention weight when queried with Scalar_B and BZ_GSE has overall low attention weight. Head 4 results for Scalar_B are similar to Head 3 whereas BZ_GSE has low attention weight when queried with Scalar_B and moderate weight for querying with BZ_GSE

Discussion and Conclusions

The disturbance storm time (DST) index is an important factor useful for characterizing geomagnetic storm properties. These values can be predicted before with help of machine learning and deep learning models by analyzing the solar wind parameters. Understanding the features used in model training is important. In this paper, we present feature analysis of the DST Transformer to understand model performance based on the solar parameters. Our study has provided us with good insights on the features and model behaviour.

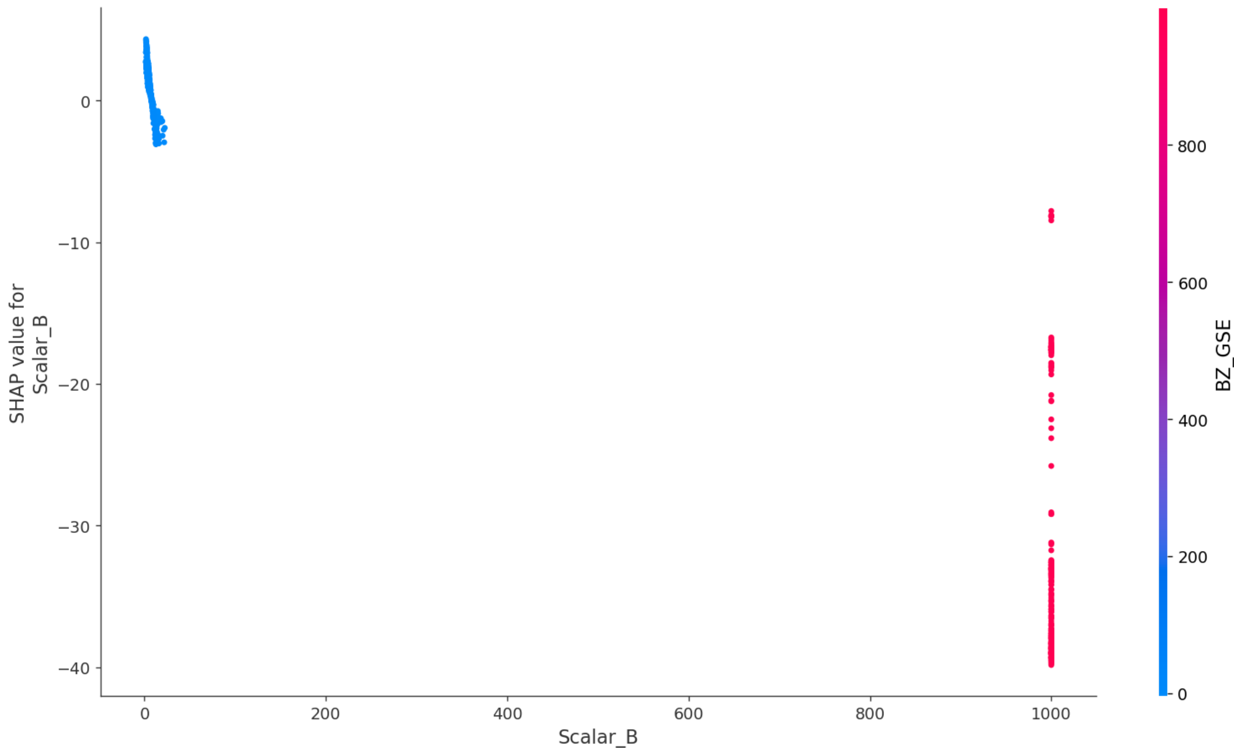

Figure 3: SHAP Dependency Plot for DST Transformer

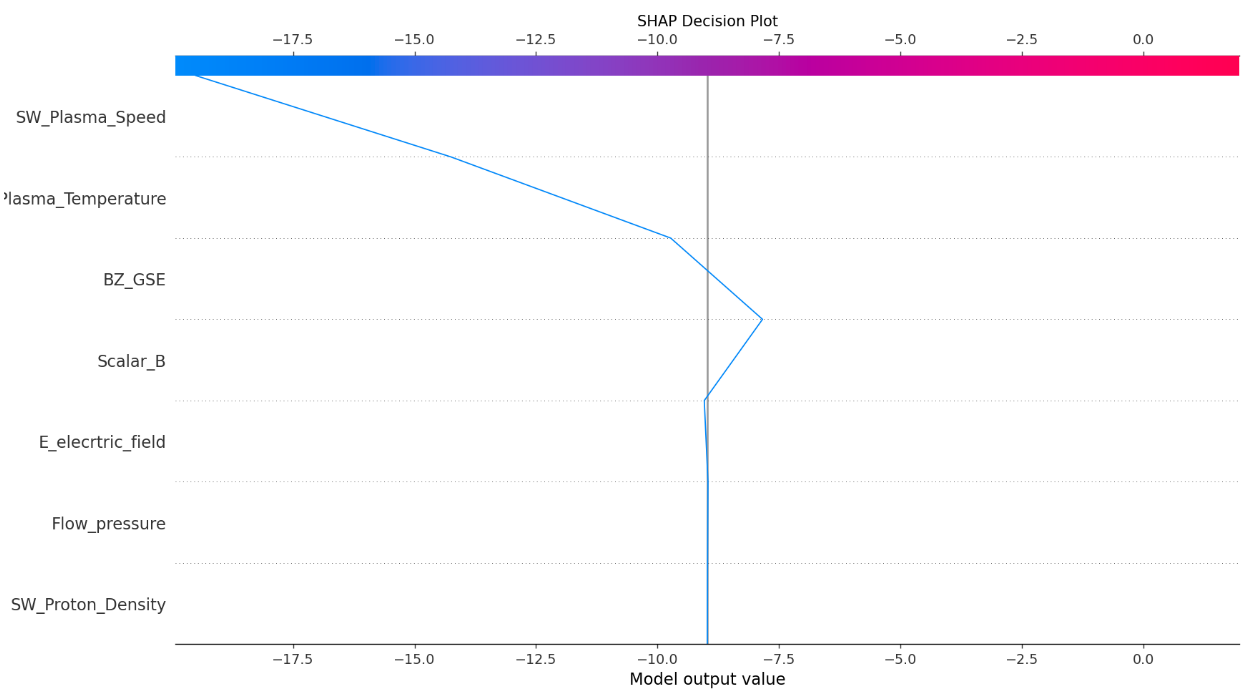

Figure 4: SHAP Decision Plot for DST Transformer

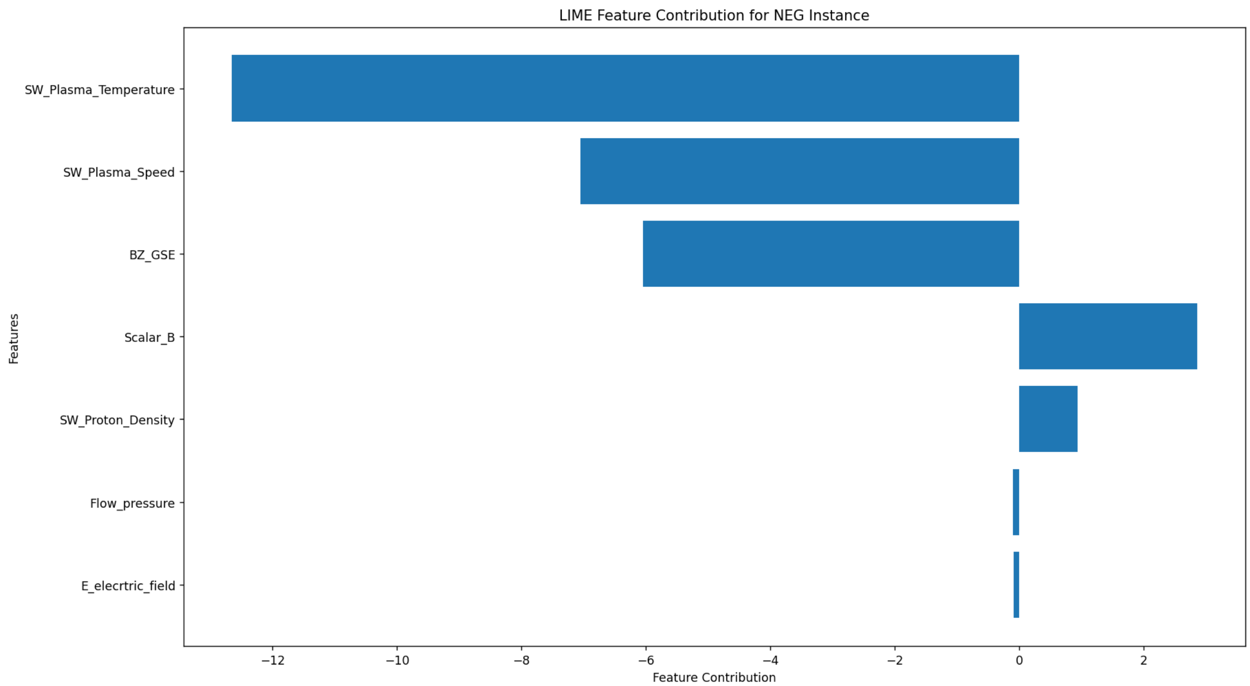

Figure 5: LIME Summary Plot for Negative Instance

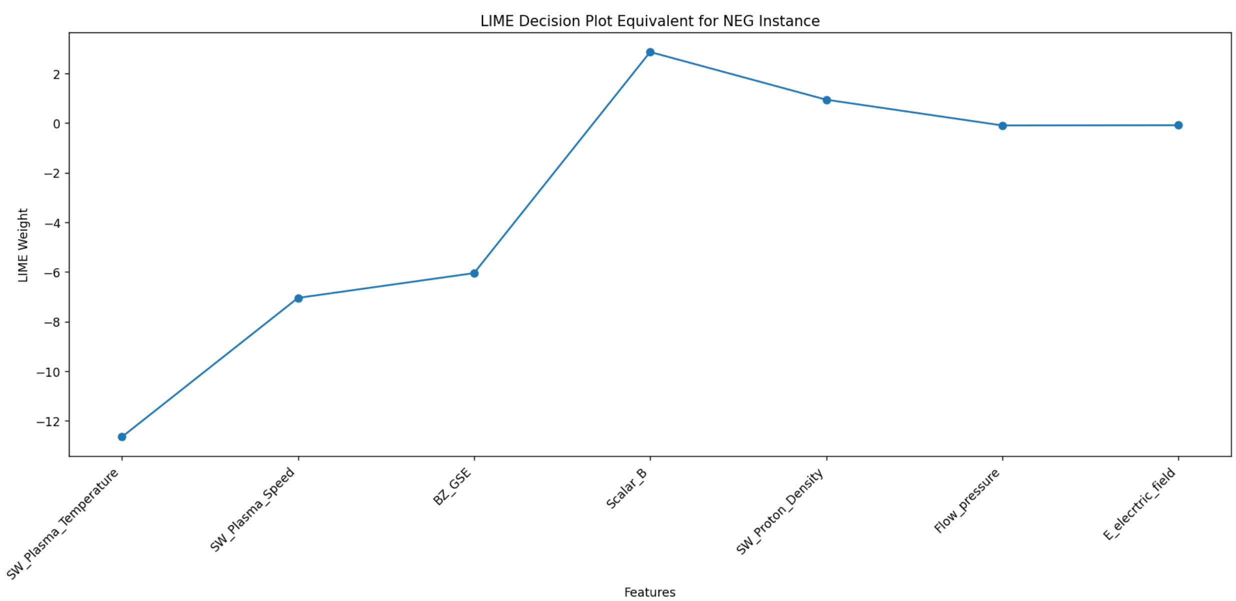

Figure 6: LIME Decision Plot for Negative Instance

The model used in our experiment is based on paper titled Forecasting the Disturbance Storm Time Index with Bayesian Deep Learning by Abduallah et al. (2022). We have also used the same data and parameters used in the paper for training and validating the model. The SHAP explainer is initialized and fed with model and the training data which is in form of cluster centers which represent the whole training data. Then shapley values are generated using test data which are plotted against test data to generate summary and dependency plots. For decision plot, the expected values from the explainer are plotted against shapley values. Then we extracted attention weights from the attention layer of

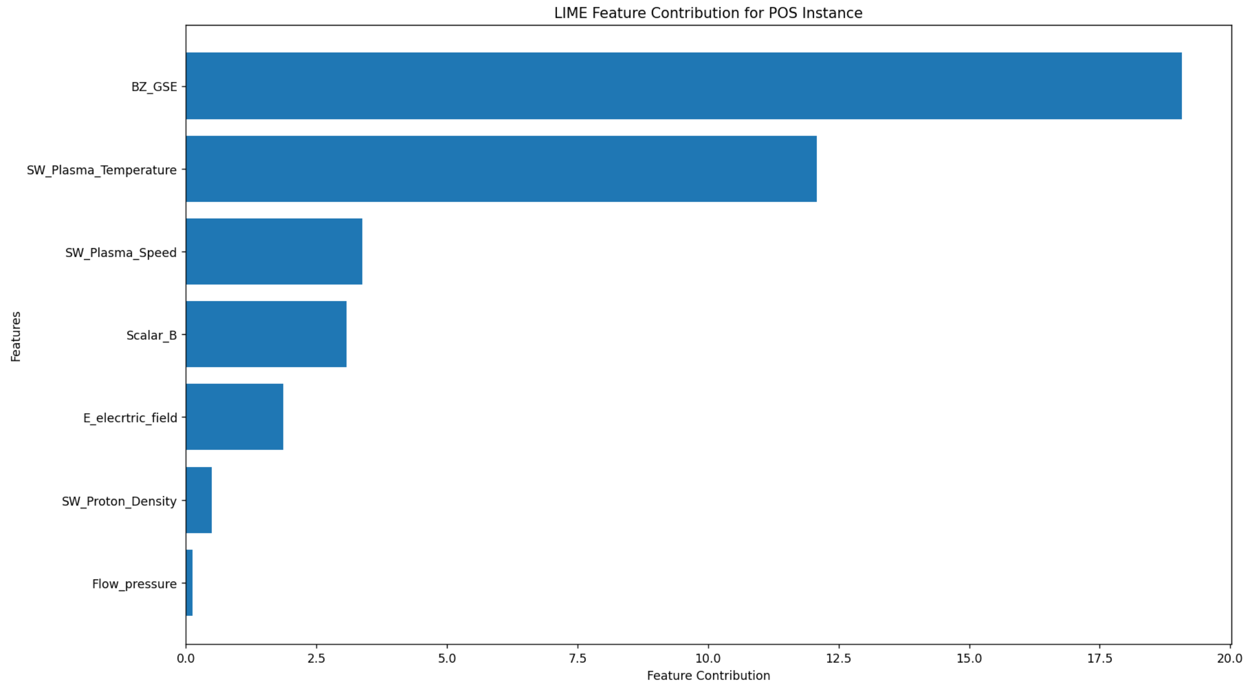

Figure 7: LIME Summary Plot for Positive Instance

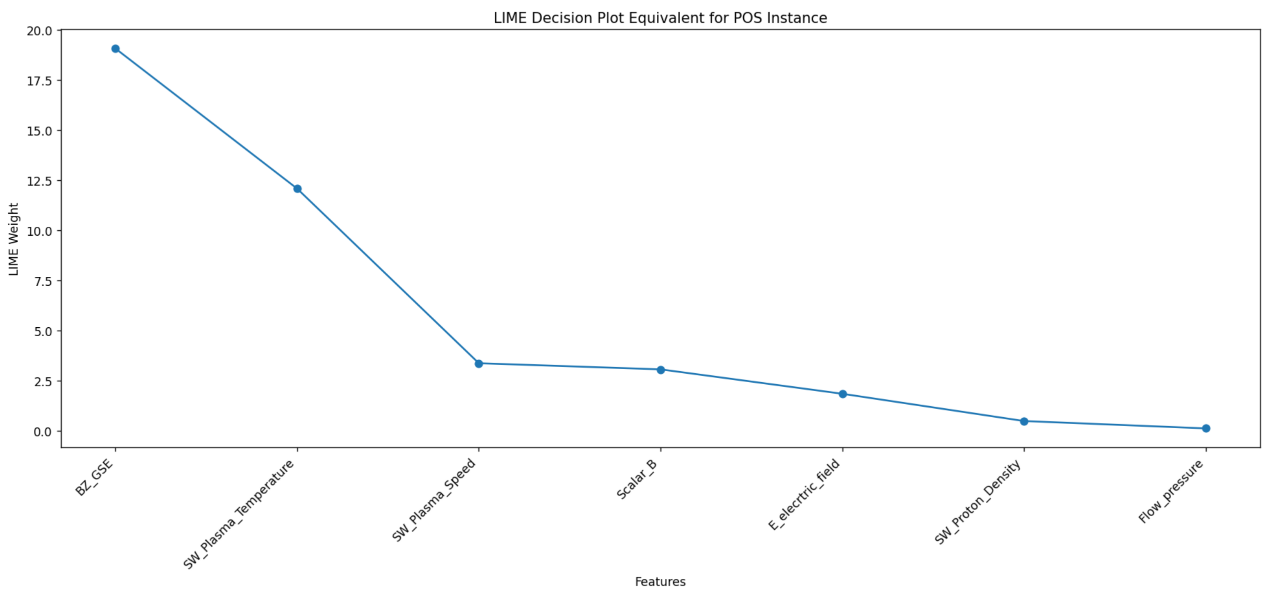

Figure 8: LIME Decision Plot for Positive Instance

the model and plotted those values using heat maps. These plots are analyzed to understand which features participate or support in model building. The features which seem not so important can be eliminated thereby reducing model complexity. From summary plot it could be understood that Scalar_B is the most important feature along with BZ_GSE feature as driving force and SW_Plasma_Temperature and SW_Plasma_Speed as supporting features. Dependency plot tell us that Scalar_B feature has major influence on the model output. As the Scalar_B value increases its SHAP value decreases stating the model output is negatively correlated with Scalar_B. This result is crucial for understanding ef-

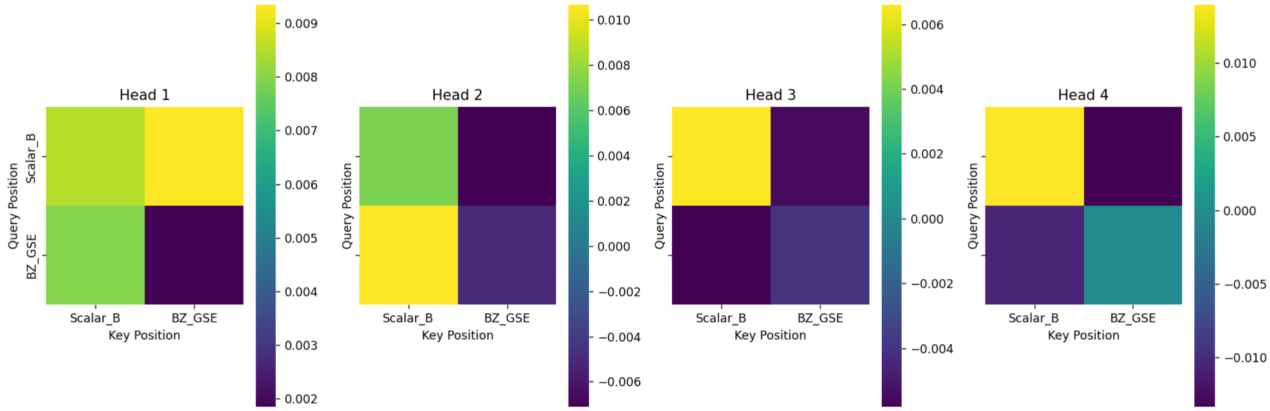

Figure 9: Attention Map for DST Transformer

fect of Scalar_B feature on the model. The plot also states that the feature BZ_GSE has good effect on the Scalar_B. From decision plot, the features SW_Plasma_Temperature and SW_Plasma_Speed can be regarded as the most influential features in driving the model results towards the stability of the model. The attention maps provide us with information that Scalar_B is an important feature across all the heads and BZ_GSE feature's attention varies across each head indicating its importance varies with context or its interaction with other features.

We have also used the LIME explainer to study the model's behavior on both positive and negative instances. The explainer is initialized using complete training data. Then, the explanations are obtained by passing one positive and one negative instance of the test sample individually through the explainer. These explanations are plotted using summary and decision plots. In case of negative instance, summary plot and decision plot illustrate that the features SW_Plasma_Temperature, SW_Plasma_Speed and BZ_GSE seem to contribute more negatively with some significant positive impact from the feature Scalar_B. For the positive instance, SW_Plasma_Temperature, SW_Plasma_Speed, BZ_GSE and Scalar_B have significant positve impact on the model output. In both cases, the impact of the features SW_Proton_Density, Flow_pressure and E_electric_field is somewhat close to zero, confirming that they are negligible. From these results SW_Plasma_Temperature, SW_Plasma_Speed, BZ_GSE and Scalar_B seem to have high influence on the model output with SW_Plasma_Temperature being critical.

From all the above results, it concluded that Scalar_B is the most important feature for the model. BZ_GSE is a critical feature by having substantial average impact and varying attention making it conditionally important. SW_Plasma_Temperature and SW_Plasma_Speed are strong influencers of the model and have substantial negative impact making it crucial for the model. The rest of the features contain smaller positive impact which can be negligible for the model. In future work, we would like further to analyze and dive deep into the model and its parameters to understand and further tune the model to reduce the complexity, increase the efficiency and stabilize the model further.

Acknowledgments

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